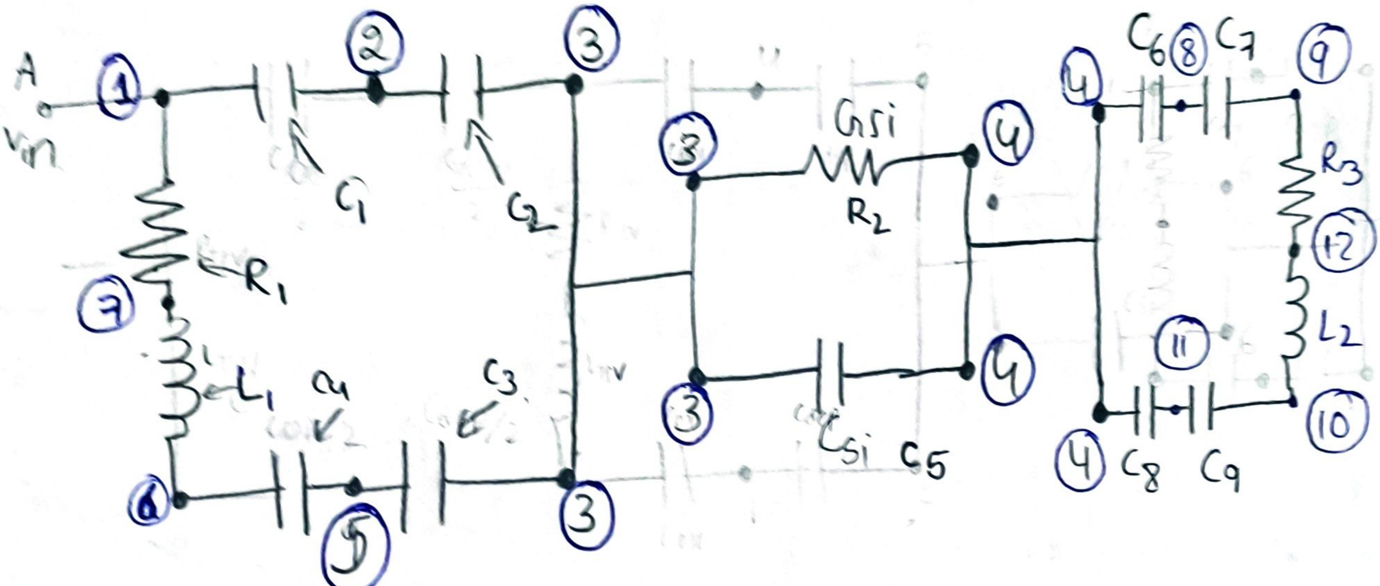
12/06/26 Day 17 Kamineni Dhana sri keerthi

Hspice (Noise, crosstalk, delay)

Defect-free Simulation



From matlab

$R_{TSV} = 8.284 \times 10^{-2}\,\Omega$	$82.8\ m\Omega$
$L_{TSV} = 2.097 \times 10^{-12} H$	$2.09\ PH$
$G_{si} = 3.540 \times 10^{-4} S$	$854\ \mu S$
$C_{si} = 3.730 \times 10^{-15} F$	$3.73\ FF$
$C_{ox} = 6.507 \times 10^{-14} F$	$65.07\ FF$
$C_{dep} = 3.584 \times 10^{-14} F$	$35.84\ FF$

Code

```
.temp25
*** Supply voltage***
Vin [illegible] 0 pulse [0 1.8 0s 100p 100p 5n 10n]
*** Network****
R1 1 7 82m
R2 3 4 354u
L1 7 6 2.0P
Cox C1 1 2 32.53f
Cdep C2 2 3 17.92f
Cdep C3 5 3 17.92f
Cox C4 6 5 32.53f
Csi C5 3 4 3.73f
R3 9 12 82m
L2 12 10 2P
dep C6 4 8 17.92f
ox C7 8 9 32.53f
dep.C8 4 11 17.92f
ox C9 11 12 32.53f.
* operation part++
.Option post
.trans
** measure ***
```

Nast delay closdallb. — later

now rise & fall time.

```
.measure tran tr trig v(1) val=0.9 rise=1 targ v(10)
+ val=0.9 rise=1
.measure tran tf trig v(1) val=0.9 fall=1 targ v(10)
+ val=0.9 fall=1
```

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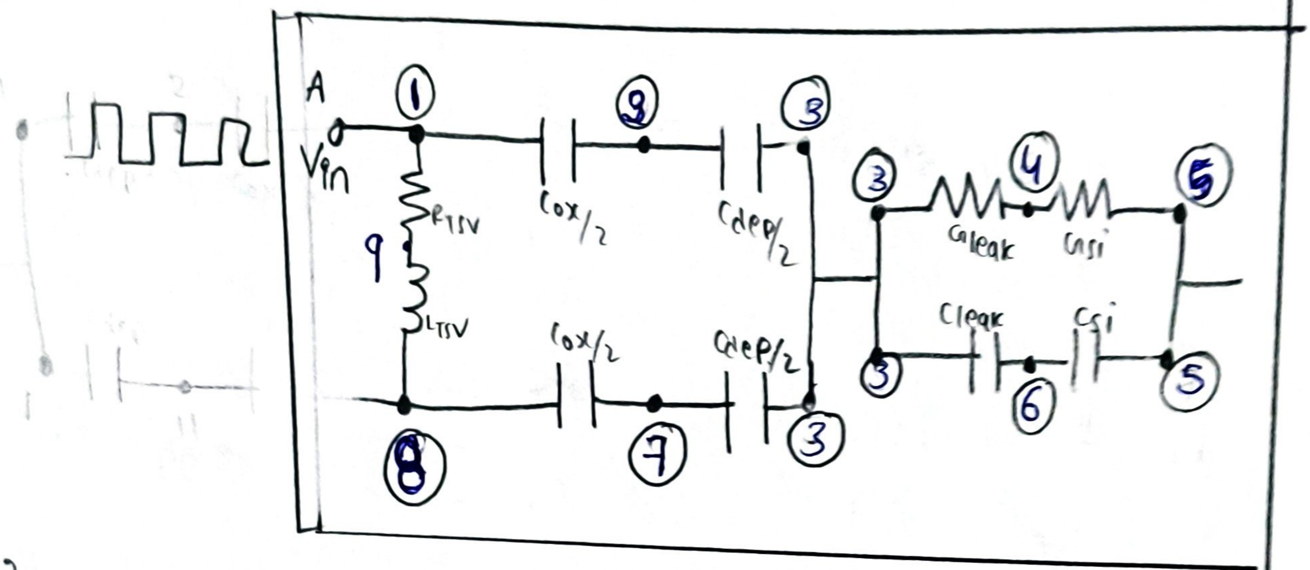
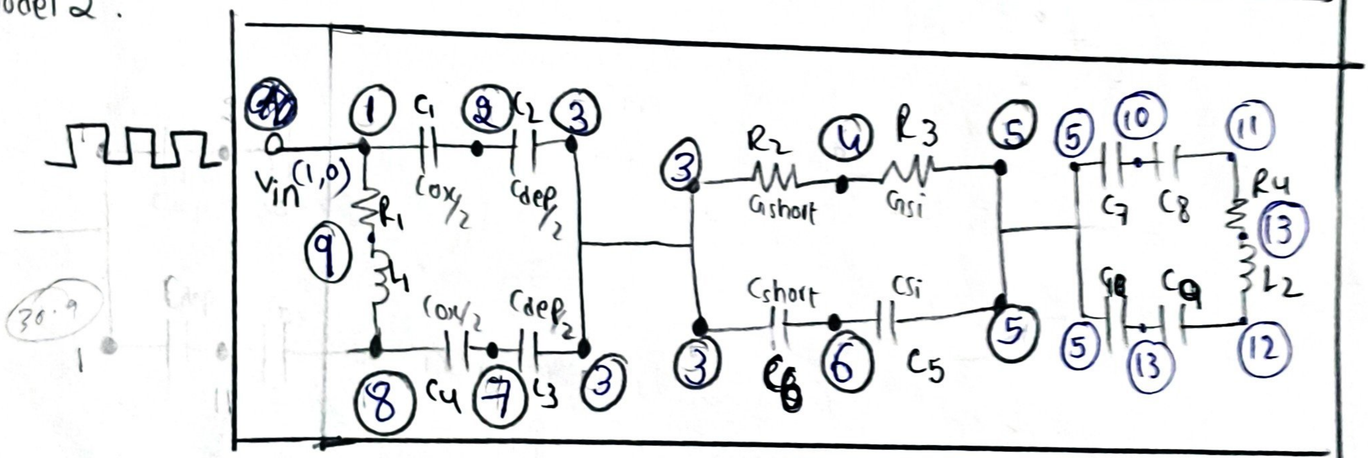
Hspice (Noise, crosstalk, delay)

Defected CG (2models (chao liu; our))

chao liu.



model 2.



Values (chaoliu)
From matlab.

$R_{TSV} = 82.8\ m\Omega$

$L_{TSV} = 2.09\ PH$

$C_{ox} = 65.0\ FF$

$C_{dep} = 34.4\ FF$

$C_{si} = 2.73\ FF$

$G_{si} = 259.8\ \mu S$

$C_{leak} = 64\ FF$

$G_{leak} = -214\ \mu S$

$82.m\Omega$

$2.09\ PH$

$C_{ox\_new} = 61.8\ FF$

Proposed

$R_{TSV} =$

$L_{TSV} =$

$C_{ox_{new}} =$

$C_{dep} =$

$C_{si} =$

$G_{si} =$

same

$C_{short} = 14.09\ FF$

$G_{short} = 108.4\ \mu S$

```
.temp 25
** supply **
Vin 1 0 pulse(0 1.8 0s 100p 100p 5n 10n)

* Network *
R1 1 9 82m
crack/break R2 3 4 214u
R3 4 5 259u
R4 11 13 82m
L1 9 8 2p
L2 13 12 2p
ox C1 1 2 30.9f
dep C2 2 3 17.2f
dep C3 7 3 17.2f
ox C4 8 7 30.9f

si C5 6 5 2.73f
leak/short C6 3 6 64f
dep C7 5 10 17.90f
ox C8 10 11 30.9f
ox C9 13 12 30.9f
dep C10 5 13 17.90f

or -> our proposed R2 3 4 108.4u
                   C6 3 6 14f

.measure tran tr trig v(1) val=0.9 rise=1 targ v(10) val=0.9
rise=1
. - - -
  - - - -
```

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Δ Simulation [Defect free, CG CL, CG OUR, along with eq], RL eq derive.

L_{sh}:- $$\frac{\mu_0 \mu_{TSV} t_{ins}}{2\pi}\left[\cosh^{-1}\left[\frac{t_{ins}}{\sqrt{L_s^2+t_{ins}^2}}\right]-\sqrt{2t_{ins}^2+L_s^2}+\frac{t_{ins}}{4}+\sqrt{L_s^2+t_{ins}^2}\right]$$